

KUWAIT UNIVERSITY
Department of Mathematics, Calculus I (Math 101)
First Exam Solutions

1. [4 + 4 + 4 + 4 + 4 = 20 pts.] The graph of the function f is given. Use it to answer each of the following:

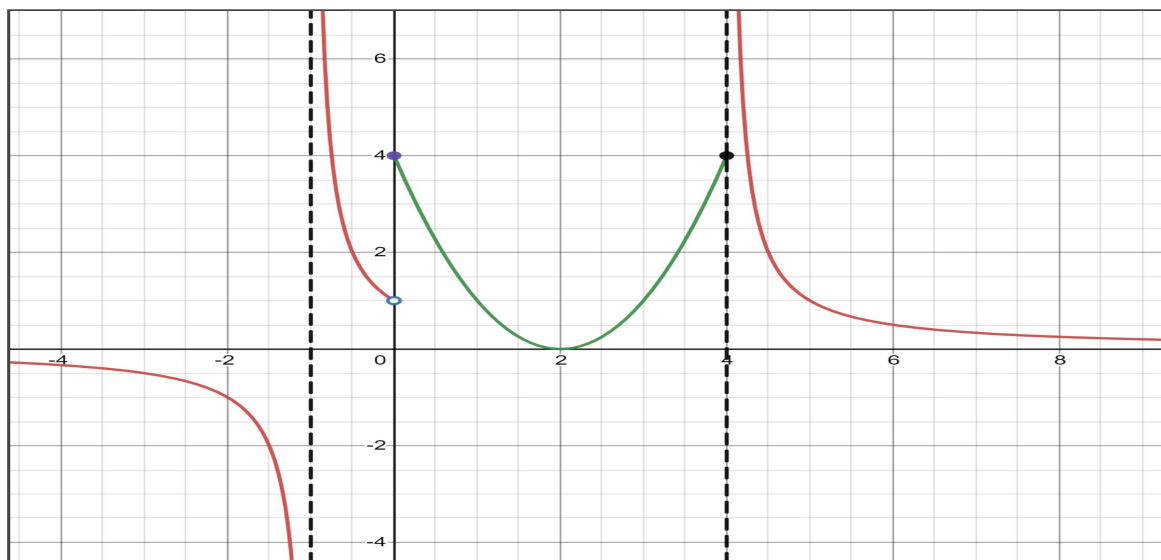


Figure 1: The graph of the function $y = f(x)$.

- (a) Find $\lim_{x \rightarrow 4} f(x)$, if it exists. Justify your answer.

We have $\lim_{x \rightarrow 4^+} f(x) = \infty$, therefore $\lim_{x \rightarrow 4} f(x)$ does not exist.

- (b) Find $\lim_{x \rightarrow 0} f(x)$, if it exists. Justify your answer.

We have $\lim_{x \rightarrow 0^+} f(x) = 4$ and $\lim_{x \rightarrow 0^-} f(x) = 1$, therefore $\lim_{x \rightarrow 0} f(x)$ does not exist.

- (c) Find all vertical and horizontal asymptotes.

We have $x = -1, 4$ are the only vertical asymptotes and $y = 0$ is the only horizontal asymptote.

- (d) Is f differentiable at $x = 0$? Justify your answer.

It is clear that it is not continuous since $\lim_{x \rightarrow 0} f(x)$ does not exist which implies that f is not differentiable at $x = 0$.

- (e) Find all points of discontinuity.

The points of discontinuity are $x = -1, 0, 4$.

2. [15 pts.] Find the domain, range and a formula for the inverse of the function $f(x) = \ln(e^{2x} - 2)$.

To find the domain, we have to have $e^{2x} - 2 > 0$ or $e^{2x} > 2$ this is equivalent to $2x > \ln 2$ or simply $x > \frac{\ln 2}{2}$ after taking the natural logarithmic function to both sides of the inequality.

Thus the domain of his function is $(\frac{\ln 2}{2}, \infty)$. Now we find the inverse by first taking $y = \ln(e^{2x} - 2)$. Next we apply the exponential function to both sides to obtain $e^y = e^{2x} - 2$ or simply $e^{2x} = e^y + 2$. This is equivalent to $2x = \ln(e^y + 2)$ or simply $x = \frac{\ln(e^y + 2)}{2}$. Therefore, the inverse is $f^{-1}(x) = \frac{\ln(e^x + 2)}{2}$. It is clear that the domain of f^{-1} is all of $\mathbb{R}$ since $e^x + 2 > 0$ for all x . In other words, the range of f is $\mathbb{R}$.

3. [8 + 7 pts.] Determine the point of discontinuity and its type of the given function.

$$(a) f(x) = \begin{cases} \frac{x^2 - x - 6}{|x + 2|}, & \text{if } x \neq -2, \\ 5, & \text{if } x = -2. \end{cases}$$

$$(b) f(x) = \frac{1}{|e^x - e^{-x}|}.$$

- (a) We have $f(-2) = 5$. On the other hand, we have

$$\lim_{x \rightarrow -2^-} f(x) = \lim_{x \rightarrow -2^-} \frac{x^2 - x - 6}{-x - 2} = \lim_{x \rightarrow -2^-} \frac{(x + 2)(x - 3)}{-x - 2} = \lim_{x \rightarrow -2^-} -x + 3 = 5.$$

Also, we have

$$\lim_{x \rightarrow -2^+} f(x) = \lim_{x \rightarrow -2^+} \frac{x^2 - x - 6}{x + 2} = \lim_{x \rightarrow -2^+} \frac{(x + 2)(x - 3)}{x + 2} = \lim_{x \rightarrow -2^+} x - 3 = -5.$$

This implies that $f(-2) = \lim_{x \rightarrow -2^-} f(x) \neq \lim_{x \rightarrow -2^+} f(x)$ which implies that f has a jump discontinuity at $x = -2$.

- (b) We have

$$\lim_{x \rightarrow 0^\pm} f(x) = \lim_{x \rightarrow 0^\pm} \frac{1}{|e^x - e^{-x}|} = \infty.$$

This implies that f has infinite discontinuity at $x = 0$.

4. [10 pts.] Find the exact value of $\sec\left(\tan^{-1}\left(\frac{2}{5}\right)\right)$.

We let $\theta = \tan^{-1}\left(\frac{2}{5}\right)$. This is equivalent to say that $\tan \theta = \frac{2}{5}$. From Pythagorean theorem, we obtain $\cos \theta = \frac{5}{\sqrt{29}}$ which implies that $\sec\left(\tan^{-1}\left(\frac{2}{5}\right)\right) = \sec \theta = \frac{1}{\cos \theta} = \frac{\sqrt{29}}{5}$.

5. [10 pts.] Evaluate the given limit, if it exists.

$$(a) \lim_{x \rightarrow 1} \frac{x^2 - 1}{\sqrt{x + 3} - 2}.$$

$$(b) \lim_{\theta \rightarrow 0} \sin^{-1}\left(\frac{1 + \tan \theta}{2 \cos \theta}\right).$$

(a)

$$\begin{aligned}\lim_{x \rightarrow 1} \frac{x^2 - 1}{\sqrt{x+3} - 2} &= \lim_{x \rightarrow 1} \frac{(x-1)(x+1)}{\sqrt{x+3} - 2} \times \frac{\sqrt{x+3} + 2}{\sqrt{x+3} + 2} \\&= \lim_{x \rightarrow 1} \frac{(x-1)(x+1)(\sqrt{x+3} + 2)}{(x+3) - 4} \\&= \lim_{x \rightarrow 1} \frac{(x-1)(x+1)(\sqrt{x+3} + 2)}{(x-1)} \\&= \lim_{x \rightarrow 1} (x+1)(\sqrt{x+3} + 2) = 8.\end{aligned}$$

(b)

$$\begin{aligned}\lim_{\theta \rightarrow 0} \sin^{-1} \left(\frac{1 + \tan \theta}{2 \cos \theta} \right) &= \sin^{-1} \left(\lim_{\theta \rightarrow 0} \left(\frac{1 + \tan \theta}{2 \cos \theta} \right) \right) \\&= \sin^{-1} \left(\frac{1}{2} \right) = \frac{\pi}{6}.\end{aligned}$$

6. [10 pts.] Use the definition of the derivative to find the derivative of the function $f(x) = x^2 + 3kx + 1$, where k is a constant.

We have

$$\begin{aligned}f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{(x+h)^2 + 3k(x+h) + 1 - (x^2 + 3kx + 1)}{h} \\&= \lim_{h \rightarrow 0} \frac{x^2 + 2hx + h^2 + 3kx + 3kh + 1 - x^2 - 3kx - 1}{h} \\&= \lim_{h \rightarrow 0} \frac{2hx + h^2 + 3kh}{h} = \lim_{h \rightarrow 0} \frac{h(2x + h + 3k)}{h} = \lim_{h \rightarrow 0} 2x + h + 3k = 2x + 3k.\end{aligned}$$

7. [10 pts.] Find an equation of the tangent line to the graph of $f(x) = \frac{x^2 + 2}{(x+1)e^x}$ at $x = 0$.

We have $f'(x) = \frac{((x+1)e^x)(2x) - (x^2+2)((x+1)e^x + e^x)}{((x+1)e^x)^2}$ using the quotient and product rules for differentiation. Now $f(0) = 2$ and $f'(0) = -4$. Thus an equation of the tangent line is given by $y - 2 = -4(x - 0)$ or simply $y = -4x + 2$.

8. [10 pts.] Use the Intermediate Value Theorem to show that the function $f(x) = x^3 - 5 \sin x + 7$ has at least one x -intercept.

We have $f(0) = 7 > 0$ and $f(-3) = -27 - 5 \sin(-3) + 7 = -20 - 5 \sin(-3) < 0$. The function f is continuous on the closed interval $[-3, 0]$ as it is the difference of two continuous functions, a cubic polynomial and a scalar multiple of a trigonometric function. Thus by the Intermediate Value Theorem, there exists $c \in (-3, 0)$ such that $f(c) = 0$ or equivalently $c^3 - 5 \sin c + 7 = 0$ which implies that c is an x -intercept for the given function.